Mean Price in a Lot-Based Inventory Simulation with Cheapest-First Issuing

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Abstract

We describe a discrete-event simulation of a single-item inventory with (s, S) replenishment in which stock is held in lots with individual unit prices and demand is served from the cheapest lot first. The mean price of the items currently in stock is tracked after every event. We explain how this quantity is computed and show that, under cheapest-first issuing, it is a biased and therefore unusable indicator of purchase prices: expensive lots are served last, accumulate in the stock, and push the mean price towards the upper end of the lot-price distribution.

1 Model

The simulation is implemented in Python with SimPy and consists of three components.

Stock. The stock is a set of lots. Each lot j has a quantity q_j and a single unit price p_j shared by all of its items. Lots are kept in a min-heap ordered by p_j.

Demand. Demand orders arrive as a Poisson process with rate λ, i.e. with exponentially distributed inter-arrival times. The size of an order is $\lceil X \rceil$ with $X \sim \mathcal{N}(\mu, \sigma^2)$, bounded below by 1. An order is served from the cheapest lot first; when that lot is empty the next cheapest lot is used. Demand that cannot be served is counted as lost.

Replenishment. We use an (s, S) policy with $s = S\,(1 - \tau)$, where τ is the fraction by which the stock may be empty before a re-order is triggered. After every demand order, the inventory position (on hand plus on order) is checked. If it is at or below s, an order of size $S -$ position is placed. It arrives after an exponentially distributed lead time as *one* lot whose unit price is drawn from $\mathcal{N}(\mu_p, \sigma_p^2)$ (truncated at 0.01).

2 Computation of the Mean Price

Let $L(t)$ be the set of lots in stock at time t. The number of items and their total value are

$$N(t) = \sum_{j \in L(t)} q_j, \qquad V(t) = \sum_{j \in L(t)} q_j\, p_j, \tag{1}$$

and the mean price per item in stock is

$$\bar{p}(t) = \frac{V(t)}{N(t)}. \tag{2}$$

The two sums are not recomputed from scratch. They are maintained incrementally:

Symbol	Meaning	Value
T	simulated time	800
λ	demand arrival rate	2.0
$\mu,\ \sigma$	order size mean / std. dev.	5.0 / 2.0
S	order-up-to level (capacity)	1600
τ	re-order trigger fraction	0.25
s	re-order point $S(1-\tau)$	1200
	mean lead time	5.0
	initial stock / initial price	1400 / 15.0
$\mu_p,\ \sigma_p$	lot price mean / std. dev.	10.0 / 5.0
	random seed	42

Table 1: Parameters of the simulation run shown in this paper.

- when a lot of quantity q and price p arrives: $N \leftarrow N + q$ and $V \leftarrow V + q\,p$;
- when k items are taken from the lot with the lowest price $p_{\min}$: $N \leftarrow N - k$ and $V \leftarrow V - k\,p_{\min}$; if the lot is exhausted it is removed from the heap and the next lot is used.

After every change of the stock the triple $(t, N, \bar{p})$ is appended to a history, so $\bar{p}$ is a piecewise-constant function of time. When the stock is empty, V is reset to zero to remove accumulated floating-point error, and $\bar{p}$ is undefined. Demand orders and replenishments, with their lot price, are logged separately.

3 Simulation Result

Figure 1 shows one run with the parameters of Table 1. All 8 719 demanded items were served (0 % lost). The panels show, from top to bottom: the stock level with the re-order point s and the capacity S; the mean price $\bar{p}(t)$; the size of each demand order; and the price of each replenishment lot, with the marker area proportional to the lot size.

4 Cheapest-First Issuing and the Upward Drift of the Mean Price

The second panel of Figure 1 shows the central observation. Lot prices are drawn around $\mu_p = 10$, yet the mean price of the stock grows to about 20 – more than twice the mean purchase price and even above the mean lot price plus one standard deviation. This is not a statistical accident but a structural consequence of the issuing rule.

Mechanism. Because demand is always served from the lot with the lowest price, a lot with price p_j is touched only after all cheaper lots in stock are exhausted. Under an (s, S) policy the stock is replenished long before it is empty (here the position never falls below roughly 970 items of 1600), so cheap lots are consumed continuously while expensive lots are never reached. Consequently:

1. a new lot with a below-average price is consumed almost immediately and leaves the stock;
2. a lot with an above-average price *stays*, and each such lot is added to the part of the stock that is never served;
3. the stock therefore becomes a "sediment" of the most expensive lots, and $\bar{p}(t)$ is dominated by them.

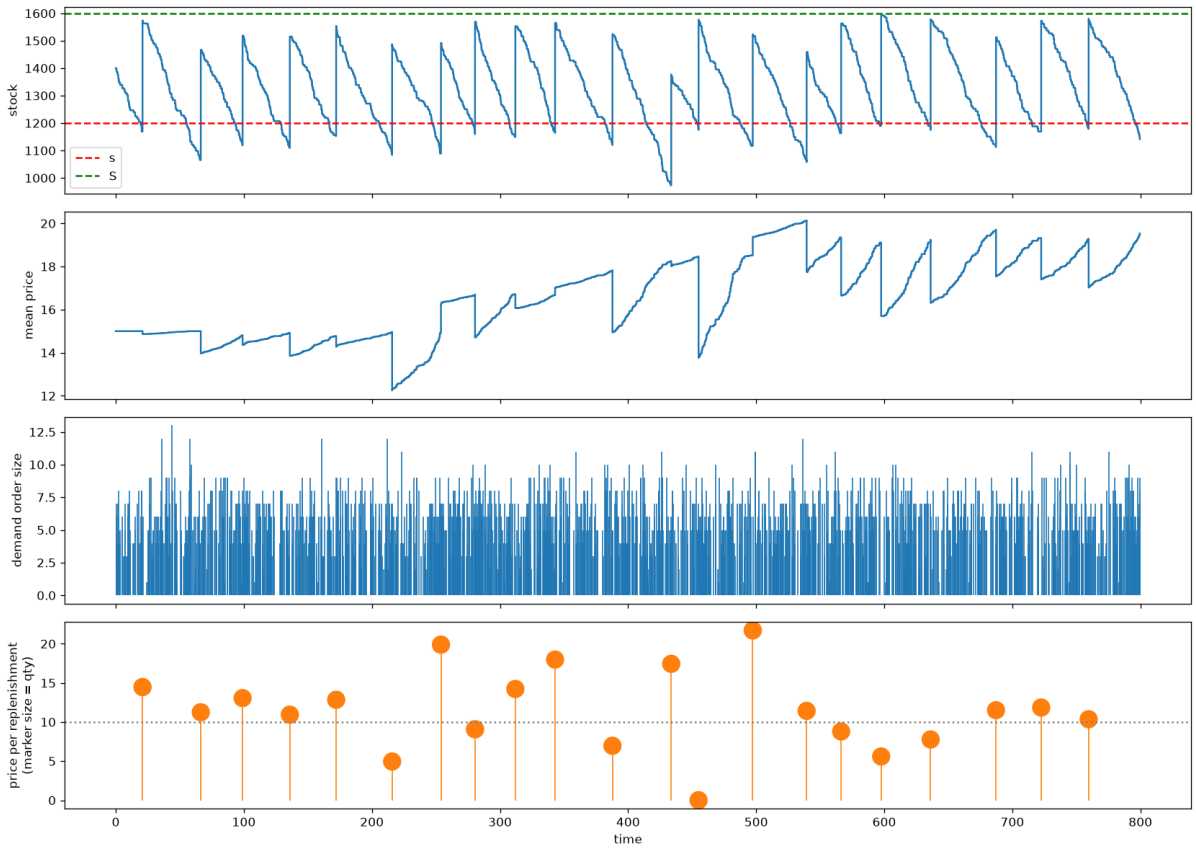


Figure 1: Simulation run. The stock level follows the typical saw-tooth of an (s, S) policy. The mean price (second panel) does not fluctuate around the mean lot price of 10 but drifts upward to roughly 20; its upward-sloping segments are interrupted only by drops when a cheap lot arrives.

The mean price rises during every period between two deliveries, because the cheapest items are removed and the average of the remaining ones increases. Each new lot lowers it only temporarily, by the amount that the lot price lies below $\bar{p}$, weighted by the lot's share of the stock.

Evidence from the run. At the end of the run the stock consists of only four lots,

Unit price	Quantity
11.85	83
17.98	249
19.88	403
21.71	406

The most expensive lot of the entire run (price 21.71, the maximum of 21 replenishments, delivered at $t \approx 497$) is still in stock with 406 items and has not been touched. Lots purchased at prices below 10 (for example 5.0, 0.0 and 5.7 in the bottom panel) were consumed soon after delivery and are not part of $\bar{p}$ any more. In effect $\bar{p}(t)$ converges towards the upper tail of the lot-price distribution. In the limit of a long run, a lot is only consumed when it is the only one left, which the replenishment policy prevents, so the most expensive lot is in practice never consumed and the mean price is pulled towards the maximum price.

5 Consequences

The mean price of the stock under cheapest-first issuing is a *survivorship-biased* quantity: it describes the items that nobody wanted to take, not the items that were bought. It has the following consequences.

- It is **not an estimator of the purchase price**. Its long-run value depends on the number of replenishments and the spread σ_p, not on μ_p.
- It is **not usable as a valuation or cost basis** for the issued items, since the issued items are by construction the cheapest ones, whereas $\bar{p}$ is determined by the most expensive ones.
- It is **not a stable state variable**: it trends with simulation length and is sensitive to the random seed through single expensive lots.
- It cannot be used to control re-ordering or pricing decisions, because it carries no information about the current market price.

For these reasons, the mean price as defined here is useful only as a diagnostic of this very bias. If a meaningful price statistic is needed, the issuing rule or the statistic must change. Options are FIFO or LIFO issuing, a moving-average cost (where issued items leave at the current mean price, so $V \leftarrow V - k\,\bar{p}$), or recording the price of the *issued* items rather than of the items in stock.

6 Conclusion

We presented a lot-based inventory simulation with (s, S) replenishment and cheapest-first issuing, and derived the incrementally maintained mean price $\bar{p} = V/N$. The simulation shows that this measure inherits a systematic upward bias from the issuing rule: expensive lots are never consumed, remain in stock, and drive the mean price towards the maximum lot price. The problem is therefore not the mean price itself but the decision to always serve from the cheapest lot. The resulting property is an artefact of the model and cannot be used as a price indicator.

Possible fix. A natural remedy is to serve demand from both ends of the price range, i.e. to take items from the cheapest *and* from the most expensive lot instead of from the cheapest lot only (for example alternately, or by splitting each order between the two). Expensive lots would then leave the stock as well, and the mean price of the stock should move back towards the mean lot price instead of towards its maximum. We have not implemented or tested this rule yet; it is the next step of this work.

Limits of the fix. Even if the mean price is brought back to the mean, this does not guarantee a profit. Items taken from the most expensive lots are sold with a low margin, or at a loss, if the selling price does not follow the purchase price. A balanced mean price therefore solves the bias of the statistic but opens a second problem, the profitability of the issuing rule.

Lesson learned. The mean price of the stock is not only an output of the simulation. In reality it influences the selling price and with it the demand. In the model presented here, demand is independent of the price: the arrival rate λ and the order size are constants. A realistic model must therefore close this loop and describe demand as a function of the (mean) price, for example $\lambda = \lambda(\bar{p})$. Only then can different issuing rules be compared by their effect on sales volume and profit and not only by the mean price they produce.